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Meta-analysis on effects of artificial intelligence education in K-12 South Korean classrooms

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Abstract
This study aimed to integrate the results of prior studies on the effectiveness of AI education in K-12 Korean classrooms to draw systematic and comprehensive conclusions. To achieve this goal, a review of 64 studies on AI education in Korea that were conducted from 2019 to 2023 was subjected to a meta-analysis. The total effect size of AI education and the effect size for each categorical variable were calculated. The results of the meta-analysis are as follows. The overall effect size of AI education is 0.897 ($95\%CI = 0.781 \sim 1.013$, $p < .05$), which can be interpreted as a large effect. In terms of learning effect, the cognitive domain had a large effect, and the affective domain had a medium effect. In addition, the effect size by publication type, school grade, subject, class type, activity type, and other categories was presented. In the early stages of AI education policy, this meta-analysis presents evidence-based action recommendations by synthesizing the results of prior studies. This study provides implications for instructional design and supportive plans that can increase the effectiveness of AI education.

Keywords Artificial intelligence education · Meta-analysis · Learning effect · Effect size · K-12

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1 Introduction

In the age of intelligent information, AI attracts attention as a game changer in many areas such as industry, economy, and culture (Makridakis, 2017). The infinite scalability of AI maximizes productivity and efficiency in various industries and has a great influence on the way we live. For this reason, there is an emphasis on everyone having AI competencies (Miao & Holmes, 2021), many countries are making efforts to develop AI-related industries and cultivate AI talent.

In particular, the role of K-12 is growing to develop artificial intelligence-related competencies (Sanusi et al., 2023). AI education aims to improve core competencies required for future society by understanding the concepts and principles of AI (Holmes et al., 2019). Teaching the basic concepts of AI had been mainly done in universities, but AI education has recently increased at the K-12 level with the emergence of tools that can easily explain concepts such as block programming languages (Marques et al., 2020). K-12 teachers teach the concepts, principles, influences, and ethics of AI, and students learn while designing, developing, and experimenting with algorithms, machine learning, and deep learning to solve complex problems. AI education is an extension of computer science (Wong et al., 2020), but new approaches are teaching AI in a way that solves complex problems by fusion with various subjects. And in countries such as the United States, China, and Republic of Korea, content standards or curriculum for AI education are being established and taught to students (UNESCO, 2022).

AI education contributes to improving the learning effectiveness of learners' cognitive and affective domains (Rizvi et al., 2023). In the cognitive area, learning effects have been reported AI understanding through unplugged activities and education using Google Teachable Machine was changed (Kim et al., 2020a), improved computational thinking skills using block coding and physical computing (Lee & Kwon, 2022), and increased data literacy through project classes (Kim & Han, 2022). In the affective domain, the following learning effects are observed. Technological attitudes change through machine learning models using machine learning for kids (Lee, 2019), changes in attitudes toward SW (software) and improvement of creative problem-solving skills through maker education using Arduino and AI robots (Hong et al., 2020). Currently, AI education is in its early stages, and experimental research is mainly conducted to verify effectiveness. In particular, in Korea's K-12 education, research on AI education is increasing quantitatively, and various learning effects are being suggested. Individual studies that have verified this effectiveness are contributing to accumulating knowledge about AI education.

Meanwhile, it is necessary to synthesize individual studies in order to provide implications for teaching and learning and a basis for establishing policies based on these educational effects. To date, AI education has not been reported much in academic journals around the world, which limits the ability to synthesize its effects. However, in Korea, AI education has been actively implemented as a policy since 2020, and a lot of individual research has been accumulated. Collecting these individual studies to determine the effect size can greatly help policymakers and researchers make decisions. With this systematic analysis method, meta-analysis can be used to calculate the effect size by synthesizing several studies (Glass, 1976), and the difference in effect size can be found based on the categorical variables. Results of effective interventions can be objectively derived from studies conducted using a variety of research methods to help with evidence-based decision-making. Additionally, meta-analysis has a post-hoc nature, so rather than providing definitive conclusions about relationships, it provides implications on the type and direction of research needed in the field (Borenstein et al., 2009). In particular, by synthesizing the effects of AI education experienced in Korea, it will be possible to provide insight into the establishment of AI education policies in other countries.

Accordingly, this study aims to provide implications for the implementation of AI education by examining the effect size of AI education. Specific research questions are as follows.

Research Question 1. What is the overall effect size of AI Education?
Research Question 2. What is the effect size according to the categorical variable of AI Education?

2 Literature review

2.1 Artificial intelligence education in K-12

With the development of AI, there is a lot of interest in how to teach and use AI in the educational field. Currently, AI education in K-12 education is divided into learning about AI and learning with AI.

First, learning about AI means teaching students AI concepts, principles, algorithms, and problem solving (Holmes et al., 2019). Representatively, Association for the Advanced Artificial Intelligence and Computer Science Teachers Associate in the United States, through AI4K12 (https://ai4k12.org/), contributed to learning content for teaching AI to K-12 students. They formed a guideline based on five big ideas in AI: perception, representation and reasoning, learning, natural interaction, and societal impact (Touretzky et al., 2019). The guideline is divided into grades K-2, 3 ~ 5, 6 ~ 8, and 9 ~ 12, and include learning objective of enduring understanding. In Korea, in 2020, focusing on the Korean Association of Information Education, Korea's next-generation SW curriculum standard was announced to include surveys, forums, workshops, and public hearings on SW education (Kim et al., 2020b). The educational content for AI is presented in Table 1.

Second, learning with AI refers to using AI as a tool to promote learning in various subjects (Holmes et al., 2019). Holmes et al. (2019) divided using AI into system-facing AI, student-facing AI, and teacher-facing AI. One the system-facing, AI can be used for admissions, timetables, learning management, attendance records, and predication of students dropout and failure. On the student-facing, AI supports customized education, such as the intelligence tutoring system and exploratory learning. There is an exploratory learning environment and an automated writing evaluation system,

Table 1 'Artificial Intelligence' area among contents of the Korean Software Education Standard Model (Kim et al., 2020b, p.354)

Stage	Contents for Artificial Intelligence (AI)
Elementary School Stage 2	- Application Cases of AI (e.g. Voice Recognition, etc.)
Elementary School Stage 3	- Understanding Concepts of AI - Application of AI (Machine Learning etc.)
Middle School Stage 1	- Understanding Knowledge an Expression of AI - Understanding Methods of Machine Learning
Middle School Stage 2	- Understanding AI Reasoning - Understanding Approach Methods of Machine Learning
Middle School Stage3	- Knowledge Reasoning in Multiple Areas - AI Learning in Multiple Areas
High School Stage 1	- Concept and Algorithm of Machine Learning - Data Analysis Methods of Machine Learning
High School Stage 2	- Concept and Programming of Artificial Neural Network - Real Life Problem Solving of Artificial Neural Network

and teachers can use it for automatic grading, student monitoring, and assistant teachers. Recently, with the emergence of ChatGPT, and Large Language Models (LLMs), efforts are being made to actively adopt them in language education (Gao et al., 2024). Because the context in which learning about AI and learning with AI are applied to teaching and learning are different, this study conducted a meta-analysis using only research papers dealing with learning about AI as the main research topic.

2.2 Key features of AI education

As standards for AI education have been proposed, the incorporation of AI education into K-12 education has increased since 2019 (Marques et al., 2020). The characteristics of AI education implemented on the K-12 level are as follows.

The content of AI education can be largely divided into AI understanding, AI experience, and AI problem-solving. First, understanding AI consists of finding definitions, concepts, ways to use AI, and ethics. Exploring the concepts of AI can include the use of AI in daily lives and discussion (Lee, 2021c), or unplugged activities (e.g., predicting pictures, finding patterns), or Convolution Neural Network (CNN, an image recognition algorithm of artificial intelligence) (Ryu & Han, 2019). In addition, in recent years, it has become important to examine the ethical use and social impact of AI (Ali et al., 2019; Borenstein & Howard, 2021; Furey & Martin, 2019). Since AI is a self-learning and evolving technology, it is necessary to consider ethical issues such as fairness, accountability, transparency, bias, autonomy, agency, and inclusion (Holmes et al., 2021). In this regard, in K-12, it is important to guide transparency, accountability, and information protection practices for the use of AI (Byun, 2020). In current K-12 classes, various classes are conducted that examine the social impact of AI and discuss ethical use (Choi & Park, 2021; Lee, 2021c).

Second, AI experience consists of using products and services to which AI is applied, or simply applying AI algorithms using a commercial platform. As a specific example, one can use the Google Teachable Machine to distinguish between dogs and cats, or to understand AI algorithms through activities such as facial recognition (Kim et al., 2020a).

Third, AI problem solving consists of the process of analyzing the cause of the problem using data and finding a solution using machine learning and artificial neural networks. As machine learning can be easily implemented with block-based programming, Python, etc., training cases using machine learning in K-12 have been increasing since 2019 (Marques et al., 2020). In Park et al.'s (2020) study, a class was conducted about making an artificial intelligence chatbot in a camp for elementary school students and parents. The researchers developed a software instructional model consisting of problem recognition and analysis, data collection, data preprocessing and feature extraction, machine learning model training and evaluation, machine learning programming, application and problem-solving, and process of sharing and feedback (Moon & Lee, 2019). In addition, a study by Moon and Lee (2019) proposed a teaching and learning method that allows beginners to easily learn the perceptron algorithm, a kind of artificial neural network, through block-based programming. Problem-solving education using AI is actively being conducted among K-12 students in the form of STEM education and maker education (Chung & Shamir, 2020; Vartiainen et al., 2020).

Various activities are taking place related to AI education. First, AI education has been conducted through unplugged methods that do not use special teaching aids. AI principles, concepts, and algorithms are taught to students using puzzles, board games, and play without utilizing computers (Lindner et al., 2019). This type of education is mainly provided to students in lower grades and is useful in conveying basic concepts about AI.

Second, block-based programming languages and text-based programming languages are used to develop AI algorithms. Tools for block-based programming: Machine learning for kids (https://machinelearningforkids.co.uk/), Teachable Machine (https://teachablemachine.withgoogle.com/), Scratch (https://scratch.mit.edu/), Entry (https://playentry.org/), etc. are being used. Classes are being conducted in which elementary school students create models that classify plant leaves using a teachable machine (Shin & Shin, 2021), or develop models that respond to user questions using machine learning for kids and Scratch (Lee, 2019). Text-based programming tools include Python, Microsoft Azure, and Google Colab. Since text-based coding is more difficult than block-based coding, it is mainly applied to middle and high school students (Jang & Yoon, 2020). Learning experiences using these tools can help learners understand AI (Sanusi et al., 2023).

Third, hardware such as sensors, Arduino, and robots are being used for AI education. Students learn the principle of machine learning by connecting ultrasound, RGB module, sound module, etc. to Arduino to collect and process external data, or to create a robot that finds its own way using programming based on AI blocks (Hong et al., 2020).

Many attempts have been made in AI education in integrated education such as STEAM education. (Sakulkueakulsuk et al., 2018; Zafari et al., 2022). Since AI technology itself has the versatility to be used in various academic fields, classes are being conducted in the form of convergence of several subjects in K-12. Related research includes media art program development using data and artificial intelligence for elementary and middle school students (Yoon et al., 2019), as well as Korean language, mathematics, society, science, physical education, art, ethics, and practical arts for elementary school students. AI comprehension education conducted in practical arts and subjects, Korean language education, basic mathematics, and more must be learned in the lower grades of elementary school (Lee, 2021c). Furthermore, materials that can be experienced through play have been developed (Song, 2020).

AI education is implemented in the form of clubs, camps, and e-learning conducted by universities and institutions outside the school as well as within schools. Lee (2020) developed an AI education program and applied to 52 middle school students for one night and two days. This study reported that AI education applying the design thinking process has a positive effect on the value perception and efficacy of AI. In addition, with the spread of COVID-19 in 2020, the Busan Metropolitan City Office of Education has allowed teachers and students to learn AI anytime. The B-MOOC (https://bmooc.pen.go.kr/) system has been developed to provide 90 courses. The contents of the lectures provided by B-MOOC include basic statistics, big data, machine learning, and Python. Students can take courses on their own, or teachers can open classes and run courses together.

The target audience for AI education in K-12 is diverse, ranging from elementary school students to high school students. For elementary school students, education on AI concepts, principles, and ethics continues using Unplugged or EPL. AI-related coding and problem-solving activities are being conducted for middle and high school students, and physical computing such as Arduino is also being actively implemented. Because the level of students varies depending on the school level, it is necessary to find customized content and activities.

Additionally, several studies have found AI education has a positive learning effect (Choi & Park, 2021; Shamir & Levin, 2020). AI education research mainly evaluates learning effects and attitudes toward AI, as shown in Table 2 (Marques et al., 2020; Su et al., 2022). First, several studies applying a program for AI learning

Table 2 Key Effects of K-12 AI education studies

Key effects	Related studies
Understanding on AI	Choi and Park (2021), Hong and Kim (2020), Kim et al. (2020a), Salas-Pilco (2020), Shamir and Levin (2020)
Attitude toward AI	Burgsteiner et al. (2016), Hong et al. (2020), Han (2020b), Kim et al. (2020a), Lee (2021c), Lee et al. (2021b), Lee (2020), Salas-Pilco (2020), Shamir and Levin (2020), Shin and Shin (2021), Vachovsky et al. (2016), Lee (2019b)
Recognition of AI Related careers	Han (2020b), Vachovsky et al. (2016)
Computational thinking	García et al. (2020), Hong et al. (2020), Shamir and Levin (2020)
Problem solving	Hong et al. (2020), Lee (2021c), Yoon et al. (2019)

have suggested that there is a positive effect on cognitive areas such as the concept of AI, how to use AI, principles of machine learning, and understanding of algorithms. In a study by Choi and Park (2021), a tool that can easily teach the K-NN algorithm to elementary school students was applied to the class, and the principles of machine learning, understanding an algorithm, and applying machine learning were positively improved per a pre-and post-test methodology. A study by Shamir and Levin (2020) used interviews to confirm that students understand AI processes after AI education for elementary school students. Hong and Kim (2020) emphasized the cultivation of data literacy in AI education, and suggested that AI education has positive effects in all areas of data understanding, collection, analysis, and expression.

Second, many studies have suggested that AI education has a positive effect on affective areas such as interest, efficacy, confidence, and motivation for AI. Vachovsky et al. (2016) reported that the Stanford Artificial Intelligence Laboratory's outreach summer program for high school girls was improved confidence in AI and Computer Science. In addition, Shin and Shin (2021) found that after three weeks of AI education among elementary school students, perception of AI changed to familiarity, novelty, convenience, diversity, and safety.

In addition, a number of studies have shown improvement in AI-related career consciousness (Han, 2020b), computing thinking (Shamir & Levin, 2020), problem solving skills (Hong et al., 2020; Lee, 2021c). Currently, the importance of AI education is increasing in Korea, and it is expected that studies examining the educational effect of primary and secondary education AI will continue (Han & Shin, 2020).

Several studies have been conducted on the effectiveness of AI education in the past two to three years. While individual studies encourage policy decision-making (Van den Bergh et al., 2013), a meta-analysis of multiple studies can reveal the effects of AI education comprehensively and scientifically. Categorical analysis can be used to determine which treatment is more effective in AI education and to make decisions related to policy direction or education practice. This study can be used as a basis for preparing evidence-based practices or policies for education in AI.

2.3 AI education policy in Korea

In December 2019, various government ministries in Korea jointly announced the *National Strategy for Artificial Intelligence* (Korean Government, 2019). The government established a vision of '*Beyond an IT powerhouse to become an AI powerhouse*' with nine strategies and 100 targets in three fields (building a world-leading AI ecosystem, a country that best utilizes AI, and implementing people-centered AI). It worked on this strategy, which proposed activating AI convergence in school education and lifelong education so that all citizens can use AI well, and to expand SW and AI education opportunities to strengthen computer thinking skills of K-12 students. In addition, it was announced that it would prepare educational programs for each life cycle and job group so that all citizens could cultivate digital literacy. In line with the national strategy, the Ministry of Education announced the "Directions and Core Tasks of Education Policy in the

AI Era in November 2020" in cooperation with relevant ministries (Korean Government, 2020). The Ministry of Education established the vision of the "realization of an educational paradigm in which humanity and futureness coexist" and presented the core tasks: (1) Education that focuses on humans, (2) Education that fits the times, and (3) Education that combines with technology. Detailed tasks according to the core tasks are shown in Table 3. The Presentation included the introduction of AI education into school curriculum in stages, the operation of leading schools (500 schools in 2021), the establishment of AI basics in high schools, the establishment of AI mathematics courses, and the operation of an AI convergence education in graduate schools to enhance teacher competency (38 graduate schools, 5,000 students, from 2021 to 2025), customized teaching-learning system development, educational system development for the vulnerable in education, crisis detection and prevention system development, and 4th generation development of work efficiency.

With a national emphasis on AI capabilities, the provincial offices of education are also actively implementing AI education policies as shown in Table 4. The AI education policies characteristic of each provincial office of education are as follows. Seoul Metropolitan City Office of Education (2021) expands data and AI-based problem-solving learning in various subjects, develops careers in new AI technologies, trains 1,000 AI education experts, develops customized AI training solutions, and utilizes AI tutors for vulnerable groups. Busan Metropolitan City Office of Education (2021) signed an MOU with companies, research institutes, and universities to cooperate with each other in developing AI education programs, holding educational events, and strengthening teacher competency. Incheon Metropolitan City Office of Education (2020) has developed two kinds of artificial intelligence textbooks for middle and high schools. Chungcheongnam-do Office of Education (2021) developed educational materials for advanced SW learning and educational materials for subject convergence AI education.

Table 3 Direction and key tasks of AI educational policy in the Korean Ministry of Education (Korean Government, 2020)

Task	Specific tasks
Education that focuses on people	· Concentrate School Education on Fostering Self-Directed Attitudes · Strengthening Human Dignity Education for the People-Centered AI Era
Education in line with the times	· Advance AI Education as a Future Education Step by Step · Enhance Pre-service and In-service Teachers' Competency · Educate AI Field Experts · Support for Re-training for Field Experts
Education that combines technology	· Promotion of Three Intelligent Education Projects: Learner-centered Environment, Support for the Vulnerable in Education, Student Safety and School Work Efficiency · Facilitate Data-driven Student Support Programs at Universities · Composition and Operation of the Educational Big Data Committee

Table 4 AI education policies of key Provincial Office of Education in Republic of Korea

Province	Key AI educational policy
Seoul Metropolitan City Office of Education (2021)	· Mid- to long-term development plan for AI based convergence innovation future education (2021–2025): Innovation of public education through A-based convergence education, Encouraging AI-based customized education and bridging education gap, AI-based super-personalized education environment creation. · 1,000 Professional in-service teachers and 200 leading teachers' group for AI Education (~ 2025)
Busan Metropolitan City Office of Education (2021)	· MOU to spread the value of AI education: Enterprises, Research Institutes, Universities, etc. · Launching 90 online courseware service for professional education in understanding AI and AI convergence areas.
Incheon Metropolitan City Office of Education (2020)	· Incheon AI Education Development 3 Year Project (2020–2022): AI talent cultivation, Program development and environment construction for AI education. · Textbook development for secondary school level AI
Chungcheongnam-do Office of Education (2021)	· Comprehensive Promotion Plan for Chungnam AI Education (2021–2025): Building a foundation for AI education, strengthening teachers' competencies for AI education, Operating AI curriculum, and Creating communication and sharing culture for AI. · Development and dissemination of textbook for elementary Software education in-depth learning and 8 books for curriculum convergence for Chungnam AI education

The main characteristics of the AI education policy implemented by the Korean Ministry of Education and Provincial Offices of Education are summarized as follows. First, in kindergartens and elementary schools, education on understanding and application of AI based on play and experience is emphasized, and education in understanding and solving principles is emphasized in middle and high schools. In kindergarten and elementary school levels, play or unplugged activities centered on understanding and using AI are being conducted to increase interest in technology and support self-directed learning. And in middle and high school, classroom activities are being conducted to solve problems related to real life using AI services or machine learning. Second, it emphasizes AI-based convergence education that can be applied in all subjects. The emphasis is on convergence education that solves problems in various ways using AI in all subjects, not programming-centered education that is conducted only in specific subjects. Third, the AI education competencies of in-service teachers are improved by utilizing graduate school in AI convergence education and teacher training. Graduate Schools of AI Convergence Education are open to everyone, regardless of the in-service teacher's field, and a certain portion of the tuition is covered by the Office of Education. In addition, AI education capabilities are being reinforced through various teacher training and online content. Fourth, AI-based programs and services are being developed to provide customized education considering individual levels and interests. The government is making efforts to provide customized education by building a learning management system that connects big data and AI. The private level is developing AI-based products and services that consider students' interests and levels. Fifth, the government is exploring ways to improve school safety and work efficiency and making efforts to build a safe environment by recognizing various risk factors in advance, such as intelligent CCTV and air quality management using Internet of Things. In addition, AI teachers and Robotic Process Automation are introduced to simplify and automate schoolwork, thereby attempting to reduce the administrative burden on teachers.

Korea's AI education policy is focused on strengthening the AI capabilities of all citizens. In K-12, we plan to implement AI education more systematically by introducing a revised curriculum starting in 2025. Accordingly, practice and research on AI education in K-12 will increase explosively. By providing evidence of the effectiveness of AI education through meta-analysis, it will be possible to provide a basis for policy promotion.

3 Method

3.1 Data

In order to conduct a meta-analysis on the effectiveness of AI education, related studies in academic journals, dissertations, and research conference reports published in Korea were analyzed. Academic journals and dissertations were searched in the academic information search service (http://www.riss.kr), the representative academic database in Korea, and the Korean journal citation index (https://www.kci.go.kr/). Research conference reports were searched on Edunet T-Clear (https://www.edunet.net/), which provides all research conference reports published in Korea. The report of the research competition was verified by experts as excellent in the research procedure and method during the review process, and the report that won the national competition was targeted. For data search, search terms of "Artificial Intelligence or Machine Learning" and "Education or Learning or Teaching" were used. The process of selecting the paper to be analyzed went through the process of a PRISMA flowchart as shown in Fig. 1.

As a result of the search, a total of 7,175 documents were collected. In the case of Korea, most of the papers were registered in the academic information search service and the Korean journal citation index, so there were no papers collected outside the database. Excluding 3,367 duplicate studies in all papers, 3,610 studies not related to AI education for K-12, 5 studies without main text of the manuscript, and 115 studies that did not calculate an effect size such as qualitative studies, correlation studies, and comparative studies, 8 studies with poor research validity, 6 studies with dissertation published in academic journals, 64 studies can be subjected to meta-analysis. A convenience sample of studies was finally selected. Since this

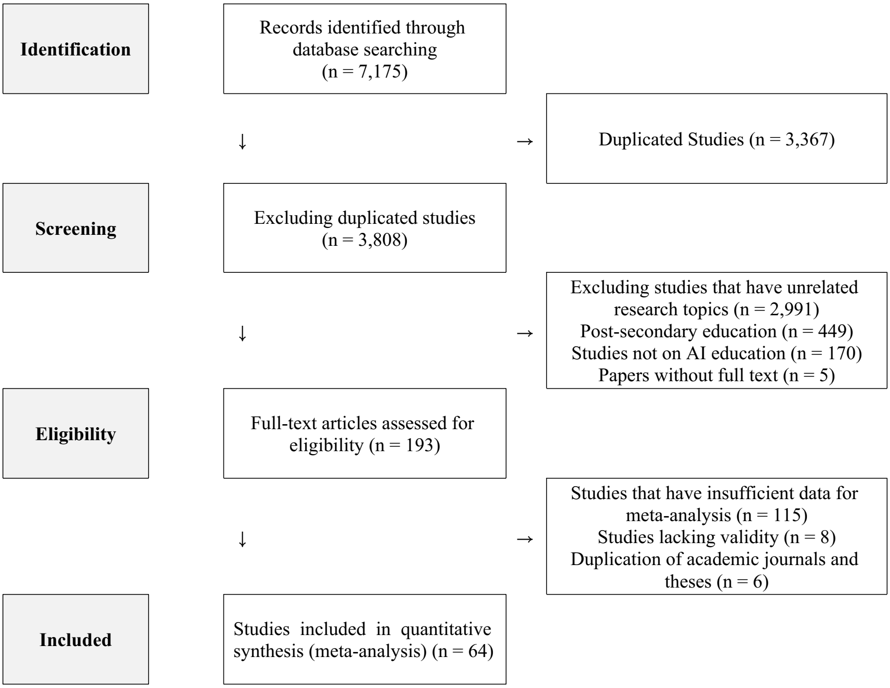


Fig. 1 PRISMA flowchart

study is to find out the magnitude of the effect of AI education, only studies that examined AI itself were selected. Research related to AIED, which uses AI as an auxiliary means of learning, such as chatbots, was not included in the final analysis. If the report and thesis of the research conference were revised or supplemented and published in an academic journal, the journal was selected for analysis. The reason for this is more valid because the journal has gone through a process of peer review or expert review. On the other hand, researchers selected only theses for which the experimental design and measurement tools of the study were valid.

3.2 Analysis method

3.2.1 Data coding

64 papers selected as subjects were reviewed, classified into categorical variables, and coded as shown in Table 5. The coding of the data was conducted by two in-service teachers who held a Ph.D. in education. To clearly distinguish each variable, when the dissertation was published as an academic journal, the instructional and learning guidance attached to the dissertation was in-depth analysis and coding. Contents that could not be confirmed during the analysis process were answered by phone calls or e-mailed questions to the corresponding author of the study. In

Table 5 Coding table for data analysis

Variable	Sub-category	Coding contents
Categorical variable	Educational effects	•Cognitive domain (Understanding AI, AI problem-solving, Understanding algorithm, Science inquiry ability, Computing thinking, etc.) •Affective domain (Attitude toward AI, SW attitude, SW self-efficacy, Technological attitude, interest, etc.)
	Publication type	•Journal papers (Korean Citation Indexed journals) •Dissertation (Degree papers and Thesis)
	School grade	•Elementary school (K-6) •Middle school (7–9) •High school (K-10-12)
	Curriculum	•Curricular subject (within school curricular subjects) •Extra-curricular (Activities such as club, camp, etc. out of curricular activities)
	Integration	•Single subject (implementation in a subject) •Integration among school subjects (implementation in multiple subjects)
	Instruction type	•Online instruction •Offline instruction •Blended (Online and offline instruction)
	Activity type	•Unplugged (Using tool(s) without using computer) •EPL (Block or text based educational programming language) •Physical computing (Using robot, Arduino, etc.)
	Contents type	•AI concept (Definition, application, ethics, social impact, etc. for AI) •AI hands-on (Utilization AI product, service, and simple algorithms) •AI problem-solving (application of machine learning and artificial neural network for problem-solving) •All integrated for concept, hands-on, and problem-solving

Fig. 2 Formula for Hedges's g

$$J=[1-\frac{3}{4(n_1+n_2)-9}]\text{or}\ (1-\frac{3}{4df-1})$$

$$d=\frac{\overline{X_1}-\overline{X_2}}{S_p}$$

$$g=J\times d$$

Table 6 Interpretation of effect size

Small effect size	Medium effect size	Large effect size
≤0.20	0.20 ~ 0.80	≥0.80

the case of disagreement during the analysis, the code was decided through consultation. For the coded data, the effect size was calculated using Comprehensive Meta Analysis version 3.3. In data coding, gender and EPL (educational programming language) type were also coded, but since all studies were targeting hybrid and block-based programming, the result was excluded from the description.

3.2.2 Calculation of effect size

In this study, the effect size was calculated as the corrected standardized mean difference (Hedges's *g*). The standardized mean difference (Cohen's d) tends to overestimate the effect size when the sample is small, and Hedges's *g* is needed to correct it (Hedges & Olkin, 1985). Since studies with small and large sample sizes are mixed, it is appropriate to calculate Hedges's *g*. The formula to find Hedges's *g* is shown in Fig. 2.

Since the sample size of each study is different, weighting is required to properly calculate the effect size (Hwang, 2016). In general, the larger the sample, the more accurate it is, so the more weight is given. In this study, weights were assigned according to the number of cases according to the method of Hedges and Olkin (1985). The effect size according to the meta-analysis result follows Table 6, which is an interpretation criterion suggested by Cohen (2013).

3.2.3 Homogeneity test

Borenstein et al. (2009) proposed a homogeneity test in meta-analysis to confirm the assumption that the individual research results to be analyzed are from the same population. As a result of the homogeneity test, the effect sizes extracted from the study subjects were found to be heterogeneous ($Q=419.524$, $p<.05$, $I^2=84.983$).

Therefore, a random-effects model was applied to the total effect size. Subgroup analysis and meta-regression analysis were applied to the effect size according to each category.

3.2.4 Verification of publication convenience

Since studies that derive positive and statistically significant results are more likely to be published than studies that do not, publication bias may exist (Oh, 2002). As a result of visually examining the funnel plot to find out whether the publication is appropriate or not, it was found that the left and right sides were not symmetrical as shown in Fig. 3. The initial value of the regression equation had to be statistically significant ($p < .05$) because of examining Egger's regression equation. This indicates that there is some publication convenience. Therefore, to quantitatively evaluate the degree of publication convenience, the stability coefficient (fail-safe N) of Rosenthal (1979) was calculated. At this time, since the coefficient (N) is 5,952, which is greater than $5k + 10$ (k: number of studies), the results of this study can be interpreted as being reliable.

3.2.5 Analysis unit

In general, since many studies report more than one effect size, it is possible to violate the assumption of independence if meta-analysis is performed by overlapping the same sample. To prevent this, Cooper (2015)'s shifting unit of analysis was applied. When calculating the overall effect size, the study was used as an analysis unit, and when analyzing subgroups according to categories, the effect size was used as an analysis unit to reduce the loss of information (Cooper, 2015; Yang et al., 2018).

4 Results

4.1 Overall effect size of AI education

The meta-analysis results for the total effect size of AI education are shown in Table 7. The overall effect size is 0.897, and the 95% confidence interval is 0.781 to 1.013. By interpreting the average effect size based on Cohen (2013), AI education can be interpreted as having a large effect size. The forest plot showing the effect size and weight of each study is shown in Fig. 4. For each study, the effect size was consistently shown in a positive direction.

4.2 Effect size for each categorical variable in AI education

The effect size for each categorical variation of AI education is shown in Tables 8.

The effect size according to the educational effect was 1.060 in the cognitive domain and 0.597 in the affective domain. According to Cohen's (2013) effect size

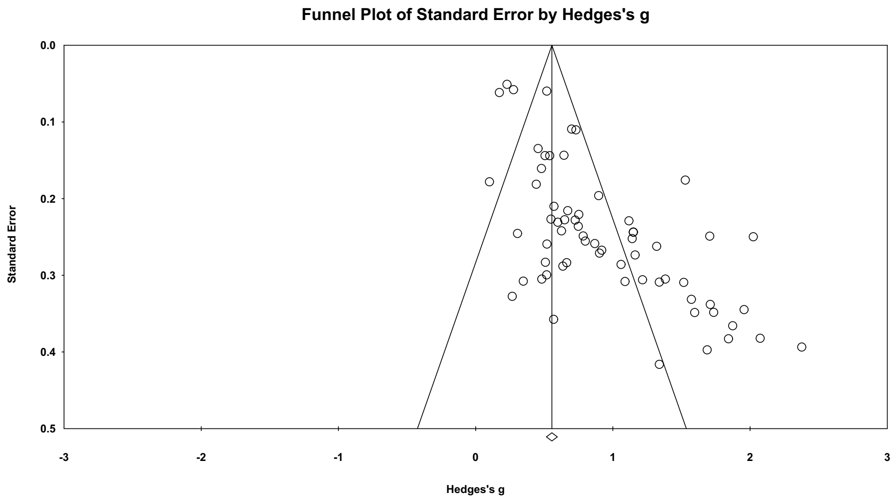


Fig. 3 Funnel plot of standard errors by Hedges's g

analysis criteria, it can be interpreted that AI education has a large effect in the cognitive domain and a medium effect in the affective domain. As a result of the analysis of variance, the significance level was $p < .05$, indicating that there was a statistical difference in the effect size according to the educational effect.

The effect size according to the type of publication was 1.046 for the thesis, and 0.684 for academic journals. Research reported in the dissertation can be interpreted as having a large effect, and studies reported in journals showed a medium effect. In the analysis of variance, the significance level was $p < .05$, indicating a statistical difference in the effect size according to the type of publication.

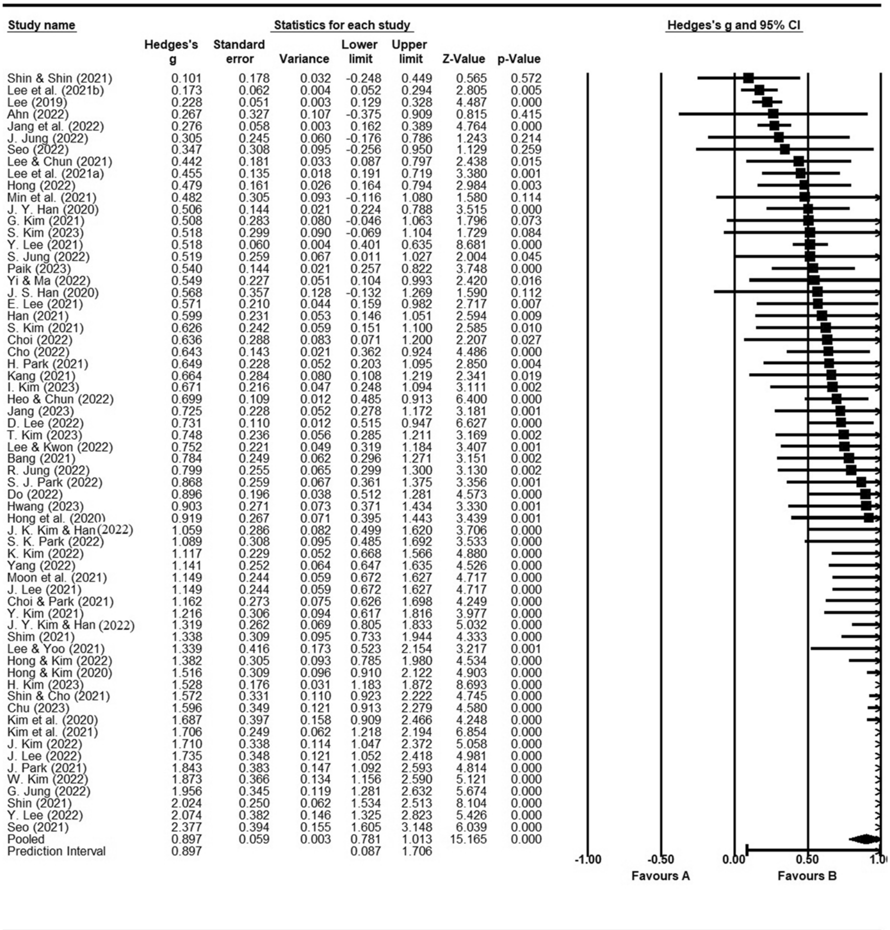
The effect size according to the school grade was 1.421 in middle schools, 0.848 in elementary schools, 0.682 in high schools, and 0.428 in the combination of elementary and middle schools. Research conducted on middle school and elementary school students can be interpreted as showing a large effect, while studies conducted on high school students demonstrated a medium effect. As a result of the analysis of variance, the significance level was $p < .05$, indicating that there was a statistical difference in the effect size according to the school grade. However, since the number of studies for middle school, high school is remarkably small, it is necessary to pay attention to the interpretation.

The effect size according to the curriculum availability was 0.911 for extracurricular activities and 0.874 for school curricular activities. It can be interpreted that AI education has a large effect size for both school curricular activities and extracurricular activities. In the analysis of variance results, the significance level was $p < .05$, indicating that there was a statistical difference in the effect size according to the subject matter.

The effect size depending on integrated education was found to be 0.929 for single course and 0.759 for integrated courses. If AI education is conducted in a single subject, it can be interpreted as having a large effect, and if it is done in an integrated

Table 7 Overall effect size

Number of studies	Homogeneity test statistic	Probability level	Effect size	95% Confidence interval	Standard error
64	419.524	< 0.05	0.897	0.781 ~ 1.013	0.059



Study name	Hedges's g	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
	Statistics for each study						
Shin & Shin (2021)	0.101	0.178	0.032	-0.248	0.449	0.565	0.572
Lee et al. (2021b)	0.173	0.062	0.004	0.052	0.294	2.805	0.005
Lee (2019)	0.228	0.051	0.003	0.129	0.328	4.487	0.000
Ahn (2022)	0.267	0.327	0.107	-0.375	0.909	0.815	0.415
Jang et al. (2022)	0.276	0.058	0.003	0.162	0.389	4.764	0.000
J. Jung (2022)	0.305	0.245	0.060	-0.176	0.786	1.243	0.214
Seo (2022)	0.347	0.308	0.095	-0.256	0.950	1.129	0.259
Lee & Chun (2021)	0.442	0.181	0.033	0.087	0.797	2.438	0.015
Lee et al. (2021a)	0.455	0.135	0.018	0.191	0.719	3.380	0.001
Hong (2022)	0.479	0.161	0.026	0.164	0.794	2.984	0.003
Min et al. (2021)	0.482	0.305	0.093	-0.116	1.080	1.580	0.114
J. Y. Han (2020)	0.506	0.144	0.021	0.224	0.788	3.515	0.000
G. Kim (2021)	0.508	0.283	0.080	-0.046	1.063	1.796	0.073
S. Kim (2023)	0.518	0.299	0.090	-0.069	1.104	1.729	0.084
Y. Lee (2021)	0.518	0.060	0.004	0.401	0.635	8.681	0.000
S. Jung (2022)	0.519	0.259	0.067	0.011	1.027	2.004	0.045
Paik (2023)	0.540	0.144	0.021	0.257	0.822	3.748	0.000
Yi & Ma (2022)	0.549	0.227	0.051	0.104	0.993	2.420	0.016
J. S. Han (2020)	0.568	0.357	0.128	-0.132	1.269	1.590	0.112
E. Lee (2021)	0.571	0.210	0.044	0.159	0.982	2.717	0.007
Han (2021)	0.599	0.231	0.053	0.146	1.051	2.594	0.009
S. Kim (2021)	0.626	0.242	0.059	0.151	1.100	2.585	0.010
Choi (2022)	0.636	0.288	0.083	0.071	1.200	2.207	0.027
Cho (2022)	0.643	0.143	0.021	0.362	0.924	4.486	0.000
H. Park (2021)	0.649	0.228	0.052	0.203	1.095	2.850	0.004
Kang (2021)	0.664	0.284	0.080	0.108	1.219	2.341	0.019
I. Kim (2023)	0.671	0.216	0.047	0.248	1.094	3.111	0.002
Heo & Chun (2022)	0.699	0.109	0.012	0.485	0.913	6.400	0.000
Jang (2023)	0.725	0.228	0.052	0.278	1.172	3.181	0.001
D. Lee (2022)	0.731	0.110	0.012	0.515	0.947	6.627	0.000
T. Kim (2023)	0.748	0.236	0.056	0.285	1.211	3.169	0.002
Lee & Kwon (2022)	0.752	0.221	0.049	0.319	1.184	3.407	0.001
Bang (2021)	0.784	0.249	0.062	0.296	1.271	3.151	0.002
R. Jung (2022)	0.799	0.255	0.065	0.299	1.300	3.130	0.002
S. J. Park (2022)	0.868	0.259	0.067	0.361	1.375	3.356	0.001
Do (2022)	0.896	0.196	0.038	0.512	1.281	4.573	0.000
Hwang (2023)	0.903	0.271	0.073	0.371	1.434	3.330	0.001
Hong et al. (2020)	0.919	0.267	0.071	0.395	1.443	3.439	0.001
J. K. Kim & Han (2022)	1.059	0.286	0.082	0.499	1.620	3.706	0.000
S. K. Park (2022)	1.089	0.308	0.095	0.485	1.692	3.533	0.000
K. Kim (2022)	1.117	0.229	0.052	0.668	1.566	4.880	0.000
Yang (2022)	1.141	0.252	0.064	0.647	1.635	4.526	0.000
Moon et al. (2021)	1.149	0.244	0.059	0.672	1.627	4.717	0.000
J. Lee (2021)	1.149	0.244	0.059	0.672	1.627	4.717	0.000
Choi & Park (2021)	1.162	0.273	0.075	0.626	1.698	4.249	0.000
Y. Kim (2021)	1.216	0.306	0.094	0.617	1.816	3.977	0.000
J. Y. Kim & Han (2022)	1.319	0.262	0.069	0.805	1.833	5.032	0.000
Shim (2021)	1.338	0.309	0.095	0.733	1.944	4.333	0.000
Lee & Yoo (2021)	1.339	0.416	0.173	0.523	2.154	3.217	0.001
Hong & Kim (2022)	1.382	0.305	0.093	0.785	1.980	4.534	0.000
Hong & Kim (2020)	1.516	0.309	0.096	0.910	2.122	4.903	0.000
H. Kim (2023)	1.528	0.176	0.031	1.183	1.872	8.693	0.000
Shin & Cho (2021)	1.572	0.331	0.110	0.923	2.222	4.745	0.000
Chu (2023)	1.596	0.349	0.121	0.913	2.279	4.580	0.000
Kim et al. (2020)	1.687	0.397	0.158	0.909	2.466	4.248	0.000
Kim et al. (2021)	1.706	0.249	0.062	1.218	2.194	6.854	0.000
J. Kim (2022)	1.710	0.338	0.114	1.047	2.372	5.058	0.000
J. Lee (2022)	1.735	0.348	0.121	1.052	2.418	4.981	0.000
J. Park (2021)	1.843	0.383	0.147	1.092	2.593	4.814	0.000
W. Kim (2022)	1.873	0.366	0.134	1.156	2.590	5.121	0.000
G. Jung (2022)	1.956	0.345	0.119	1.281	2.632	5.674	0.000
Shin (2021)	2.024	0.250	0.062	1.534	2.513	8.104	0.000
Y. Lee (2022)	2.074	0.382	0.146	1.325	2.823	5.426	0.000
Seo (2021)	2.377	0.394	0.155	1.605	3.148	6.039	0.000
Pooled	0.897	0.059	0.003	0.781	1.013	15.165	0.000
Prediction Interval	0.897			0.087	1.706		

Fig. 4 Forest plot for meta-analysis

subject, it can be interpreted as having a medium effect. As a result of the analysis of variance, the significance level was $p < .05$, indicating that there was a statistical difference in the effect size according to whether integrated education was conducted.

The effect size according to the instruction type was 0.960 for offline classes, 0.648 for blended classes, and 0.171 for online classes. It can be interpreted that AI education has a large effect when it is conducted offline classes, a medium effect

Table 8 Effect sizes by categorical variables for AI education

Variable	Sub-variables	Number for effect size	Effect size	95% Confidence interval	Standard error	Q(df)	*p*-value
Education effect	Cognitive domain	115	1.060	0.965 ~ 1.156	0.049	54.914(1)	0.000
	Affective domain	81	0.597	0.520 ~ 0.674	0.039		
Publication type	Thesis (Dissertation)	98	1.046	0.932 ~ 1.160	0.058	26.963(1)	0.000
	Journal paper	98	0.684	0.609 ~ 0.759	0.0.38		
School grade	Middle school	16	1.421	1.076 ~ 1.766	0.176	32.457(3)	0.000
	Elementary school	156	0.848	0.776 ~ 0.920	0.037		
	High school	17	0.682	0.543 ~ 0.821	0.071		
	Elementary and middle school	7	0.428	0.228 ~ 0.607	0.097		
Curriculum	Extra-curricular	25	0.911	0.725 ~ 1.098	0.095	48.412(1)	0.000
	Curricular	164	0.874	0.802 ~ 0.946	0.037		
Integration	Single	113	0.929	0.834 ~ 1.024	0.049	6.951(1)	0.008
	Integrated	83	0.759	0.675 ~ 0.842	0.043		
Instruction type	Offline	161	0.960	0.875 ~ 1.046	0.044	148.891(2)	0.000
	Blended	25	0.648	0.531 ~ 0.766	0.060		
	Online	8	0.171	0.074 ~ 0.267	0.049		
Activity type	EPL	133	0.883	0.803 ~ 0.962	0.040	8.881(3)	0.114
	EPL + Physical computing	24	0.860	0.688 ~ 1.031	0.087		
	Unplugged + EPL	18	0.832	0.621 ~ 1.043	0.108		
	Unplugged	14	0.779	0.581 ~ 0.977	0.101		
Content type	All (Concept + Hands-on + Problem-solving)	115	1.043	0.930 ~ 1.157	0.058	41.000(2)	0.000
	Problem-solving	37	0.775	0.632 ~ 0.919	0.073		
	Concept + Hands-on	44	0.585	0.502 ~ 0.668	0.042		

when it is conducted in a blended class, and a small effect when it is conducted online. In the analysis of variance, the significance level was $p < .05$, indicating that there was a statistical difference in the effect size according to the instruction type.

The effect size according to the activity type was 0.883 for EPL, 0.860 for EPL + physical computing, 0.832 for unplugged + EPL, and 0.779 for unplugged. When AI education consists of EPL, EPL + physical computing, and unplugged + EPL activities, it can be interpreted as having a large effect, and if it is composed of unplugged activities, it can be interpreted as having a medium effect. The analysis of variance found the significance level of $p > .05$, There appeared to be no statistical difference depending on the type of activity.

The effect size according to the content type was in the order of 1.043 for integration (concept + experience + problem solving), 0.775 for problem-solving, and 0.585 for concept + hands-on. When AI education content is integrated, it can be interpreted as a large effect size, and when it is problem solving or concept + experience, it can be interpreted as a medium effect size. In the analysis of variance, the significance level was $p < .05$, indicating that there was a statistical difference in the effect size according to the content type.

5 Conclusion and recommendations

5.1 Discussion

This study was conducted to find out the effectiveness of AI education conducted to date. For this, a meta-analysis was performed, and the effect size was calculated using the corrected standardized mean difference (Hedges's *g*). Key findings show that the overall effect size of AI education is large. In terms of learning effect, the cognitive domain had a large effect, and the affective domain had a medium effect. This is discussed specifically as follows:

First, the total effect size of AI education is 0.897, which can be interpreted as a large effect size. In addition, the effect sizes of individual studies used in meta-analysis were consistently shown in a positive direction. Although direct comparison is difficult as there has been little meta-analysis research on AI education, it is in line with the findings of Su et al. (2022) and Rizvi et al. (2023), which showed that AI education has positive learning outcomes through a systematic literature review. Additionally, it is consistent with reports that Computer Science education, in a similar context, has a positive effect size on learning (Scherer et al., 2020; Xu et al., 2023). This is meaningful in that it quantitatively presented a specific effect size in response to the argument that AI should be actively taught in K-12.

Second, the conclusions on the effect size of each categorical variable in AI education are as follows. In terms of educational effect, AI education has a large effect for the cognitive domain and a medium effect on the affective domain. Since the Korean government is strengthening school education so that all citizens can use AI well (Korean Government, 2019), AI education currently conducted in K-12 helps students to understand the concepts, principles, applications, and algorithms, etc. It can be interpreted as being remarkably effective. Meanwhile, to better understand

and use AI, it is important to reduce negative perceptions and attitudes toward AI (Shin et al., 2017). With technophobia, some students can recognize AI as a dangerous entity with free will, so education is required to reduce vague fears of artificial intelligence. It is important to educate students so that they can become more interested in AI, recognize that AI is playing an important role in our lives, and increase their willingness to choose AI-related jobs in the future (Han, 2020b). And currently, research in the K-12 field is focused on understanding the concept of AI (Su et al., 2022), but additional research should be conducted to develop the affective effect toward AI.

In terms of publication type, AI education has a large effect in dissertations and a medium effect in journals. In meta-analysis, publication bias is determined by comparing the effect size between published articles and unpublished articles. In general, publication bias occurs when the effect size of a manuscript published in an academic journal is larger than that of an unpublished manuscript. In this study, the opposite result is derived, which can be interpreted as the publication bias is not large.

In terms of the school grade, AI education has a large effect in studies targeting middle and elementary school students, and medium effect in studies targeting high schools.

In Korea, AI-related activity-oriented education is selectively provided in the Practical Arts subjects of elementary schools. And systematic education on SW and AI education has been provided focusing on information and technology subjects in middle school. This increase in educational opportunities for AI education can be interpreted as contributing to increasing learning effectiveness. As research results show that curriculum-based education is more effective, more research needs to be conducted on middle and high schools.

In terms of subject matter, when AI education is conducted as extra-curricular activities, it has a larger effect size than curricular activities. Extracurricular classes are operated in the form of clubs, camps, and e-learning, and are often made with the voluntary participation of students interested in AI. Such self-directed and voluntary participation can be a variable that enhances learning effects (Garrison, 1997). On the other hand, research is required to explore various variables related to the learning outcomes of AI education to enhance the learning effect in the classes within the subject.

In terms of whether convergence education, AI education has a large effect when implemented in a single subject and a medium effect when executed in a convergence subject. This can be more effective when classes are conducted in AI-related subjects in that AI education is based on understanding and programming of algorithms. However, AI education converges with a variety of subjects and, when combined with maker education or project classes, can promote various competencies such as collaboration and problem-solving skills (Hong et al., 2020). For AI convergence education to be implemented more, many studies that can give implications to instructional design need to be done.

In terms of class type, AI education has a large effect size when executed as an offline class, a medium effect size when executed as a blended class, and a small effect size when executed as an online class. This is consistent with previous studies that suggested that offline class is more effective than online class (Anstine & Skidmore, 2005; Brown & Liedholm, 2002). In addition, AI training takes place in a variety of hands-on forms, such as unplugged activities, programming, and making, but such activities may be restricted in online or blended environments. However, with the spread of Covid-19 in 2020, some schools in Korea also offer online or blended classes. In order to increase the learning effect even in such a learning environment, research on the development of teaching and learning materials and teaching methods needs to be conducted continuously.

In terms of activity type, AI education has a large effect when executed as an EPL activity, an EPL + physical computing activity, an Unplugged + EPL activity, and a medium effect when executed as an unplugged activity. Many studies have been taught around EPL, and other types of activities require careful interpretation of the results of this study because of the fact that the number of studies is small. Mainly for elementary school students, lessons were conducted around EPL using teachable machine, machine learning for kids, scratch, etc., and it was found that there was some learning effect. From the results of the study, when physical computing activities such as Arduino and sensor board are added to EPL activities, it can be seen that there is a positive learning effect. In general, physical computing activities require comprehensive programming such as device design, configuration, and control, so more in-depth learning is possible (Lee, 2019). This suggests that AI education can be effectively implemented when combined with various forms of learning such as maker education, STEAM education, and project learning (Ng et al., 2023). Additionally, because AI can be effectively taught to young learners through unplugged activities (Lindner et al., 2019), more related research and practice need to be conducted.

In terms of learning content, AI education has a larger effect than would have dealt with integrated content (concept + hands on + problem solving). AI education was systematically implemented in the aspect that most of the studies were approached with integrated content. Additionally, learning effects are more effective when approaching complex problem solving based on a basic understanding of AI.

5.2 Theoretical and practical implications

Based on these findings, the implications for AI education are summarized as follows. First, since AI education has been shown to have cognitive and affective effects at the entire school level, it is necessary to further encourage AI education in school education. Understanding the concepts, principles, and applications of AI can help develop the AI capabilities required by the future society. In addition, AI education can reduce fear of AI, create a positive attitude, and promote the correct use of AI. In particular, experience with block and text-based programming through AI education creates a positive attitude toward AI. However, there is a limit in that many studies deal only with the basic aspects of AI for elementary school students, so it is necessary to provide more in-depth AI education such as machine learning and artificial neural networks for middle and high school students.

Second, AI education needs to pay more attention to cultivating problem-solving capabilities using AI in connection with maker education or project learning, rather than simply developing programming capabilities. In this study, AI education showed the greatest effect when concept, experience, and problem solving were integrated. For these reasons, students should be presented with complex unstructured problems and encouraged to solve problems using programming, making, and physical computing from a convergent perspective of various subjects. To this end, AI education needs to be actively considered in the context of STEM education, maker education, etc. (Asunda et al., 2023). This should involve the implementation of various support strategies such as AI convergence education execution, coaching strategy, and teaching and learning case development.

Third, to increase the effectiveness of AI education, it is necessary to develop various models and explore teaching and learning strategies. In this study, AI education showed a larger effect size for extracurricular activities than for curriculum activities. In general, extracurricular activities such as clubs and camps are limited in that only a small number of students from some classes participate. In terms of ensuring universal educational opportunities, various strategies are required to increase the effectiveness of AI education in school education. In addition, AI education showed a greater effect size for offline classes than blended classes or online classes. Offline classes may be appropriate for hands-on classes such as AI education, but in order to overcome various crisis situations such as Covid-19, it is necessary to prepare a plan to increase the learning effect in blended or online classes.

Based on the research results, practical implications that can be provided to the K-12 field are as follows. First, it is crucial to develop teachers' ability to effectively guide AI education (Pedro et al., 2019). To this end, it is necessary to develop a professional development program that can identify the capabilities of teachers who provide AI education, analyze education needs, and effectively support them. It is necessary to develop AI education capabilities among computer science teachers. As AI education has recently attracted attention, computer science teachers who have already been assigned to schools have not been able to complete AI education in their pre-service time. Retraining of existing teachers needs to be carried out in stages. On the other hand, for general teachers who do not teach computer science, the ability to approach teaching and learning related topics from a holistic perspective and to solve problems using AI products or services is required. To this end, it is necessary to implement a professional development program that can effectively support the competency of teachers who conduct AI education and analyze educational needs.

Second, related support tools need to be developed so that AI education can be easily implemented in various subjects. If AI education is only focused on coding, it is difficult to implement except for by computer science teachers. In order to learn the concepts of AI in detail, programming languages such as C++ and Python will eventually be learned in subjects other than computer science subject (How & Hung, 2019). For this reason, it is necessary to develop machine learning and artificial neural network tools that anyone can easily use and distribute them to the school field. For this, it is necessary to develop related products or services based on the educational needs at the K-12 level.

Third, there is a need for research on ways to connect data, robotics, physical computing, and IoT that can be linked to AI education. AI is based on big data, but there is not enough data for AI education. In many studies, data prepared by teachers is provided randomly, or machines are trained with insufficient data. To increase the effectiveness of AI education, it is necessary to support schools by establishing a big data database for education with reliability and validity. In addition, AI has scalability to various fields such as robots, IoT, and the cloud. Teaching and learning should be conducted so that students can not only experience EPL but also experience convergence with other technologies.

5.3 Limitations and recommendations

The limitations of this study are as follows. First, there is a need to analyze the effectiveness of AI education using rigorous research methods. Among the subjects of analysis in this study, many studies have the limitation of adopting a single-group pretest-posttest design. There is a need to conduct follow-up research on studies conducted on control-experimental groups.

Second, a qualitative meta-analysis should be conducted on the effectiveness of AI education. Meta-analysis is meaningful in showing statistical results on effect size, but by adding qualitative research and presenting analysis results, the effectiveness of artificial intelligence education will be able to be comprehensively understood.

Third, research is needed on variables that affect the effectiveness of AI education. This study has a limitation in that it only investigated the effect size according to categorical variables based on experimental research. Meta-Analytic Structural Equation Modeling can be used to identify patterns or relationships by analyzing complex relationships between various variables reported in research (Cheung & Chan, 2005). This variable analysis may provide implications for the instructional design of AI-based education.

Fourth, a meta-analysis that synthesizes various types of publications is needed. Publication bias has a significant impact on research results. In this study, the publication ratio of thesis and journal papers was balanced and statistically showed no publication bias. However, if policy reports and research conference publications were also included in future studies, a more reasonable effect size could be calculated.

Appendix

Table 9 Results of coding

No	Researcher(s) (Year)	ES(g)	Publication type	Learning effect	School grade	Curricular subject	Integration	Instruction type	Activity type	Content type
1	Shin and Shin (2021)	0.10	Journal	Cognitive	ES	Curricular	Integrated	Online	EPL	All
2	Lee et al. (2021b)	0.17	Journal	Affective	ES	Not identified	Single	Online	Unplugged + EPL	All
3	Lee (2019)	0.23	Journal	Affective	ES	Curricular	Single	Offline	EPL	Problem-Solving
4	Ahn (2022)	0.27	Dissertation	Affective	ES	Curricular	Single	Offline	EPL + Physical Computing	All
5	Jang et al. (2022)	0.28	Journal	Affective	ES, MS	Curricular	Single	Not identified	EPL	Concept + Hands-on
6	Jung (2022b)	0.31	Dissertation	Cognitive	HS	Curricular	Single	Offline	EPL	All
7	Seo (2022)	0.35	Dissertation	Cognitive	ES	Not identified	Single	Offline	Others	All
8	Lee and Chun (2021)	0.44	Journal	Affective	ES	Curricular	Integrated	Blended	EPL	Concept + Hands-on
9	Lee et al. (2021a)	0.46	Journal	Affective	ES	Curricular	Integrated	Offline	EPL	All
10	Hong (2022)	0.48	Dissertation	Cognitive	MS	Curricular	Single	Offline	EPL	Problem-Solving
11	Min et al. (2021)	0.48	Journal	Cognitive	ES	Curricular	Integrated	Offline	EPL	All
12	Han (2020b)	0.51	Journal	Affective	ES, MS	Extra-Curricular	Single	Offline	EPL	All
13	Kim (2021a)	0.51	Dissertation	Cognitive	ES	Curricular	Integrated	Offline	EPL	All

Table 9 (continued)

No	Researcher(s) (Year)	ES(g)	Publication type	Learning effect	School grade	Curricular subject	Integration	Instruction type	Activity type	Content type
14	Kim (2023c)	0.52	Dissertation	Cognitive	ES	Curricular	Integrated	Offline	Others	Concept + Hands-on
15	Lee (2021c)	0.52	Journal	Cognitive, Affective	ES	Curricular	Integrated	Blended	EPL	Concept + Hands-on
16	Jung (2022d)	0.52	Dissertation	Cognitive, Affective	HS	Curricular	Single	Offline	Physical Computing	All
17	Paik (2023)	0.54	Dissertation	Affective	ES	Curricular	Single	Offline	EPL	Concept + Hands-on
18	Yi and Ma (2022)	0.55	Journal	Affective	ES	Curricular	Single	Offline	Unplugged	Concept + Hands-on
19	Han (2020a)	0.57	Dissertation	Cognitive	HS	Extra-Curricular	Single	Offline	EPL	All
20	Lee (2021a)	0.57	Dissertation	Cognitive	HS	Extra-Curricular	Single	Offline	EPL	All
21	Han (2021)	0.60	Dissertation	Affective	ES	Curricular	Integrated	Offline	Unplugged + EPL	All
22	Kim (2021b)	0.63	Dissertation	Cognitive	ES	Curricular	Single	Online	EPL	Concept + Hands-on
23	Choi (2022)	0.64	Dissertation	Affective	ES	Extra-Curricular	Single	Offline	EPL + Physical Computing	Concept + Hands-on

Table 9 (continued)

No	Researcher(s) (Year)	ES(g)	Publication type	Learning effect	School grade	Curricular subject	Integration	Instruction type	Activity type	Content type
24	Cho (2022)	0.64	Dissertation	Cognitive, Affective	MS	Curricular	Integrated	Offline	EPL	Concept + Hands-on
25	Park (2021a)	0.65	Dissertation	Cognitive	ES	Curricular	Single	Offline	EPL	Concept + Hands-on
26	Kang (2021)	0.66	Dissertation	Cognitive, Affective	ES	Curricular	Single	Blended	Unplugged + EPL	All
27	Kim (2023b)	0.67	Dissertation	Cognitive	ES	Curricular	Single	Offline	EPL + Physical Computing	All
28	Heo and Chun (2022)	0.70	Journal	Cognitive	HS	Curricular	Single	Blended	EPL	Concept + Hands-on
29	Jang (2023)	0.73	Dissertation	Cognitive	ES	Curricular	Single	Offline	EPL	Problem-Solving
30	Lee (2022a)	0.73	Dissertation	Cognitive	ES	Curricular	Single	Offline	EPL + Physical Computing	All
31	Kim (2023d)	0.75	Dissertation	Affective	ES	Curricular	Single	Offline	Unplugged + EPL	All
32	Lee and Kwon (2022)	0.75	Journal	Cognitive	ES	Curricular	Integrated	Offline	EPL + Physical Computing	Problem-Solving
33	Bang (2021)	0.78	Dissertation	Cognitive	ES	Curricular	Integrated	Offline	EPL	All
34	Jung (2022c)	0.80	Dissertation	Cognitive, Affective	ES	Curricular	Single	Offline	Unplugged	Concept + Hands-on
35	Park (2022a)	0.87	Dissertation	Affective	ES	Curricular	Integrated	Offline	Others	Concept + Hands-on

Table 9 (continued)

No	Researcher(s) (Year)	ES(g)	Publication type	Learning effect	School grade	Curricular subject	Integration	Instruction type	Activity type	Content type
36	Do (2022)	0.90	Dissertation	Affective	HS	Curricular	Single	Offline	EPL + Physical Computing	All
37	Hwang (2023)	0.90	Dissertation	Cognitive, Affective	ES	Curricular	Integrated	Offline	EPL	Concept + Hands-on
38	Hong et al. (2020)	0.92	Journal	Cognitive, Affective	HS	Extra-Curricular	Integrated	Offline	EPL + Physical Computing	All
39	Kim and Han (2022a)	1.06	Journal	Cognitive	ES	Curricular	Single	Offline	EPL	Problem-Solving
40	Park (2022b)	1.09	Dissertation	Affective	ES	Curricular	Integrated	Offline	EPL	All
41	Kim (2022b)	1.12	Dissertation	Cognitive	MS	Curricular	Integrated	Offline	EPL	All
42	Yang (2022)	1.14	Journal	Cognitive	ES	Extra-Curricular	Single	Offline	EPL	Concept + Hands-on
43	Moon et al. (2021)	1.15	Journal	Cognitive	ES	Curricular	Single	Offline	EPL	All
44	Lee (2021b)	1.15	Dissertation	Cognitive	ES	Curricular	Single	Offline	EPL	All
45	Choi and Park (2021)	1.16	Journal	Cognitive	ES	Curricular	Single	Blended	Unplugged	All
46	Kim (2021c)	1.22	Dissertation	Cognitive, Affective	ES	Curricular	Integrated	Offline	EPL	All
47	Kim and Han (2022b)	1.32	Journal	Cognitive	ES	Curricular	Integrated	Offline	EPL	All
48	Shim (2021)	1.34	Dissertation	Affective	ES	Not identified	Single	Offline	Unplugged + EPL	Concept + Hands-on

Table 9 (continued)

No	Researcher(s) (Year)	ES(g)	Publication type	Learning effect	School grade	Curricular subject	Integration	Instruction type	Activity type	Content type
49	Lee and Yoo (2021)	1.34	Journal	Cognitive	ES	Extra-Curricular	Integrated	Offline	EPL + Physical Computing	Problem-Solving
50	Hong and Kim (2022)	1.38	Journal	Cognitive	ES	Curricular	Single	Offline	EPL	Problem-Solving
51	Hong and Kim (2020)	1.52	Journal	Cognitive	ES	Curricular	Integrated	Blended	EPL	All
52	Kim (2023a)	1.53	Dissertation	Cognitive	ES	Curricular	Single	Offline	EPL	Problem-Solving
53	Shin and Cho (2021)	1.57	Dissertation	Cognitive	ES	Curricular	Integrated	Offline	EPL	All
54	Chu (2023)	1.60	Dissertation	Cognitive	ES	Curricular	Single	Offline	EPL	All
55	Kim et al. (2020b)	1.69	Journal	Cognitive, Affective	MS	Extra-Curricular	Single	Offline	Unplugged + EPL	All
56	Kim et al. (2021)	1.71	Journal	Cognitive	ES	Curricular	Single	Offline	EPL	All
57	Kim (2022a)	1.71	Dissertation	Cognitive, Affective	ES	Curricular	Integrated	Offline	EPL	All
58	Lee (2022b)	1.73	Dissertation	Cognitive	ES	Curricular	Single	Offline	EPL	Problem-Solving
59	Park (2021b)	1.84	Dissertation	Cognitive, Affective	ES	Extra-Curricular	Single	Offline	EPL	All
60	Kim (2022c)	1.87	Dissertation	Cognitive	HS	Curricular	Single	Offline	EPL + Physical Computing	All
61	Jung (2022a)	1.96	Journal	Cognitive, Affective	ES	Curricular	Single	Offline	EPL	All

Table 9 (continued)

No	Researcher(s) (Year)	ES(g)	Publication type	Learning effect	School grade	Curricular subject	Integration	Instruction type	Activity type	Content type
62	Shin (2021b)	2.02	Dissertation	Cognitive, Affective	MS	Curricular	Single	Offline	EPL	All
63	Lee (2022c)	2.07	Dissertation	Cognitive	ES	Curricular	Integrated	Offline	EPL	All
64	Seo (2021)	2.38	Dissertation	Cognitive, Affective	ES	Curricular	Single	Offline	EPL	All

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Data availability Data shared with manuscripts or supplementary information (Appendix).

Declarations

The authors have no financial or competing interests to declare that are relevant to the content of this article.

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